

Technical Design Document

Name: Adrienne

Date: June 28, 2026

Program Description: Validates a phone number, Social Security number, and ZIP code using regular expressions. The program prompts the user for input and displays whether each value is valid or invalid.

Function: `validate_phone`

Validates a phone number using a regular expression.

Parameters:

phone (str): Phone number entered by the user.

Variables:

pattern (str): Regular expression for phone validation.

Logic:

1. Receive the phone number.
2. Compare it against the regular expression.
3. Return True if valid; otherwise False.

Return:

Boolean

Function: `validate_ssn`

Validates a Social Security number using a regular expression.

Parameters:

ssn (str): Social Security number entered by the user.

Variables:

pattern (str): Regular expression for SSN validation.

Logic:

1. Receive the SSN.
2. Compare it against the regular expression.
3. Return True if valid; otherwise False.

Return:

Boolean

Function: validate_zip

Validates a ZIP code using a regular expression.

Parameters:

zip_code (str): ZIP code entered by the user.

Variables:

pattern (str): Regular expression for ZIP code validation.

Logic:

1. Receive the ZIP code.
2. Compare it against the regular expression.
3. Return True if valid; otherwise False.

Return:

Boolean

Function: main

Prompts the user for input and displays validation results.

Parameters:

None

Variables:

phone (str): User phone number.

ssn (str): User Social Security number.

zip_code (str): User ZIP code.

Logic:

1. Prompt the user for a phone number, SSN, and ZIP code.
2. Call each validation function.
3. Display whether each value is valid or invalid.

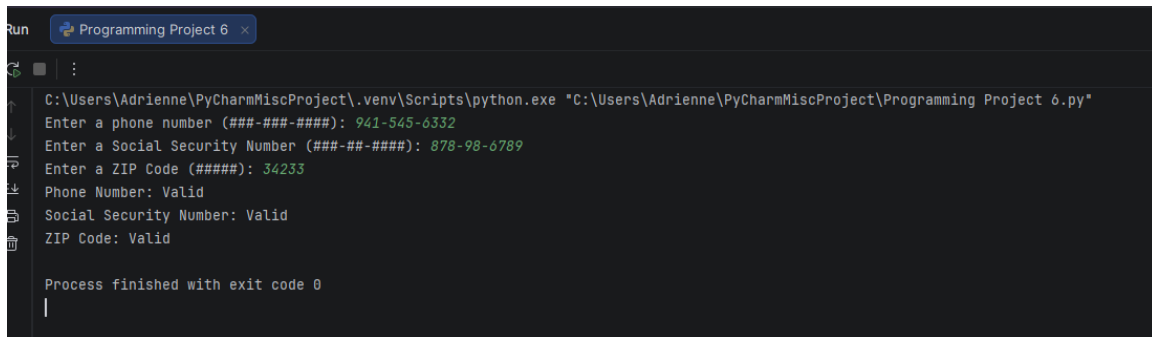
Return:

None

Repository Link

[Repos Link](#)

Output Screenshot



The screenshot shows a PyCharm Run console window for a project named "Programming Project 6". The console displays the execution of a Python script. The output shows three prompts for user input: a phone number, a Social Security Number, and a ZIP Code. Each prompt is followed by a green-colored input string. After each input, the program outputs a confirmation message: "Phone Number: Valid", "Social Security Number: Valid", and "ZIP Code: Valid". The console concludes with the message "Process finished with exit code 0".

```
Run Programming Project 6 x
C:\Users\Adrienne\PyCharmMiscProject\.venv\Scripts\python.exe "C:\Users\Adrienne\PyCharmMiscProject\Programming Project 6.py"
Enter a phone number (###-###-####): 941-545-6332
Enter a Social Security Number (##-##-####): 878-98-6789
Enter a ZIP Code (####): 34233
Phone Number: Valid
Social Security Number: Valid
ZIP Code: Valid

Process finished with exit code 0
|
```